$p$ = the probability of a success for any trial

$q$ = the probability of a failure for any trial

$p + q = 1$

$q = 1 - p$

The mean of $X$ is $\mu = np$. The standard deviation of $X$ is $\sigma = \sqrt{npq}$.

### 4.4 Geometric Distribution

$X \sim G(p)$ means that the discrete random variable $X$ has a geometric probability distribution with probability of success in a single trial $p$.

$X$ = the number of independent trials until the first success

$X$ takes on the values $x = 1, 2, 3, ...$

$p$ = the probability of a success for any trial

$q$ = the probability of a failure for any trial $p + q = 1$
$q = 1 - p$

The mean is $\mu = \frac{1}{p}$.

The standard deviation is $\sigma = \sqrt{\frac{1 - p}{p^2}} = \sqrt{\frac{1}{p}\left(\frac{1}{p} - 1\right)}$.

### 4.5 Hypergeometric Distribution

$X \sim H(r, b, n)$ means that the discrete random variable $X$ has a hypergeometric probability distribution with $r$ = the size of the group of interest (first group), $b$ = the size of the second group, and $n$ = the size of the chosen sample.

$X$ = the number of items from the group of interest that are in the chosen sample, and $X$ may take on the values $x = 0, 1, ...$, up to the size of the group of interest. (The minimum value for $X$ may be larger than zero in some instances.)

$n \leq r + b$

The mean of $X$ is given by the formula $\mu = \frac{nr}{r + b}$ and the standard deviation is $= \sqrt{\frac{rbn(r + b - n)}{(r + b)^2(r + b - 1)}}$.

### 4.6 Poisson Distribution

$X \sim P(\mu)$ means that $X$ has a Poisson probability distribution where $X$ = the number of occurrences in the interval of interest.

$X$ takes on the values $x = 0, 1, 2, 3, ...$

The mean $\mu$ is typically given.

The variance is $\sigma^2 = \mu$, and the standard deviation is $\sigma = \sqrt{\mu}$.

When $P(\mu)$ is used to approximate a binomial distribution, $\mu = np$ where $n$ represents the number of independent trials and $p$ represents the probability of success in a single trial.

## PRACTICE

### 4.1 Probability Distribution Function (PDF) for a Discrete Random Variable

*Use the following information to answer the next five exercises:* A company wants to evaluate its attrition rate, in other words, how long new hires stay with the company. Over the years, they have established the following probability distribution.

Let $X$ = the number of years a new hire will stay with the company.

Let $P(x)$ = the probability that a new hire will stay with the company $x$ years.